

Topic 2.1 - Change in Arithmetic and Geometric Sequences

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____ Class: _____ Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Describe arithmetic change as repeated addition and geometric change as repeated multiplication.
- Move among recursive, explicit, numerical, graphical, and contextual representations.
- Use discrete domains, precise notation, and contextual units.

Section A: Foundation and representation

Question 1

Core skill**No calculator**

Determine whether each sequence is arithmetic, geometric, or neither. Justify with differences or ratios: (a) 7, 12, 17, 22, ...; (b) 81, 27, 9, 3, ...; (c) 2, 6, 12, 20, ...

Question 2

Core skill**No calculator**

Write both a recursive and an explicit rule for $-9, -2, 5, 12, \dots$, using $a_1 = -9$.

Question 3

Core skill

No calculator

A geometric sequence begins $160, -80, 40, -20, \dots$. Find g_8 .

Question 4

Core skill

No calculator

An arithmetic sequence has $a_5 = -11$ and $a_{18} = 54$. Find a_{30} .

Question 5

Core skill

No calculator

A positive geometric sequence has $g_3 = 50$ and $g_7 = 4050$. Find g_1 and g_n .

Question 6

Core skill

No calculator

The table gives a sequence. Fill in the missing values and identify the invariant.

n	0	1	2	3	4
p_n	96	72	54	$\square$	$\square$

Section B: Application, comparison, and error analysis

Question 7

Modeling

No calculator

A theater has 24 seats in row 1 and each successive row has 3 more seats. Write a rule for row n , find the seats in row 18, and state the appropriate domain if the theater has 25 rows.

Question 8

Modeling

Calculator permitted

A culture contains 240 cells at hour 0 and retains 82% of its cells each hour. Write the model and determine when the population first falls below 100 cells.

Question 9

Comparison

Calculator permitted

Compare the two rules $A_n = 40 + 18n$ and $B_n = 40(1.18)^n$. Find the smallest whole-number n for which $B_n > A_n$.

Question 10

Error analysis

No calculator

A student says, "Because the common ratio is 0.72, the sequence decreases by 72% each step." Diagnose the error and write the correct percent change.

Question 11

Core skill

No calculator

Construct an arithmetic sequence whose fifth term is 9 and whose twelfth term is -12 . Then determine whether the twentieth term is positive, negative, or zero.

Question 12

Core skill

No calculator

Create a geometric sequence with $h_1 = 6$ and $h_4 = -48$. State whether the sequence alternates in sign and explain why.

Section C: AP-style multiple choice**MCQ 1**

No calculator

Which rule represents an arithmetic sequence with $a_4 = 17$ and $a_9 = -3$?

- A. $a_n = 29 - 4n$
- B. $a_n = 17 - 4(n - 1)$
- C. $a_n = 5(-4)^{n-1}$
- D. $a_n = -3 + 4(n - 9)$

MCQ 2

No calculator

A sequence is defined by $g_0 = 14$ and $g_n = 1.2g_{n-1}$. Which statement is correct?

- A. The sequence increases by 1.2 each term.
- B. The sequence increases by 20% each term.
- C. The sequence decreases by 80% each term.
- D. The sequence is arithmetic with difference 0.2.

MCQ 3

No calculator

Which observation is sufficient to prove that a sequence is geometric?

- A. The first four terms look curved when plotted.
- B. Three consecutive differences are equal.
- C. A stated rule satisfies $a_{n+1} = ra_n$ for every allowed n .
- D. Four consecutive ratios happen to be close.

MCQ 4

No calculator

The graph of a sequence is shown as isolated points. Why should the points normally remain unconnected?

- A. Connected graphs are never used in algebra.
- B. The outputs between integer indices are all zero.
- C. The sequence domain is discrete unless a continuous extension is defined.
- D. Connecting points changes the common difference.

Section D: Free-response reasoning

FRQ 1 - Coordinating representations

Calculator permitted

6 points

At week 0, Club A has 100 QAR and adds 40 QAR each week. Club B has 100 QAR and increases by 25% each week. (a) Write explicit rules. (b) Make a table for weeks 0 through 4 and identify the invariant for each sequence. (c) Compare the balances at week 8. (d) State a reasonable domain and one modeling assumption.

FRQ 2 - Evidence and limitations

Calculator permitted

6 points

A sequence has values 192, 144, 108, 81 at indices 0, 1, 2, 3. (a) Show evidence for a plausible family. (b) Write recursive and explicit rules for that model. (c) Predict the value at index 7. (d) Explain why the four values support but do not prove the continuation.
